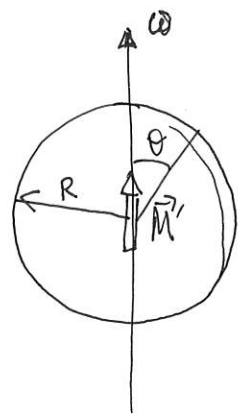
Rotating magnetic sphere (model of a pulsar)



$\vec{\omega} = \omega \hat{z}$

$\vec{M}' = M' \hat{z}$ inside the sphere.

The sphere is <u>conducting</u> (neutron star!)

Assume:

$\vec{v} = \vec{\omega} \times \vec{r}$ is small, $\frac{v}{c} \ll 1$.

$$\Rightarrow \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \approx 1.$$

In the <u>rest frame</u> of the sphere: $\vec{E}' = \vec{D}' = \vec{P}' = 0$

$\vec{\nabla} \cdot \vec{B}' = 0 \qquad \vec{\nabla} \times \vec{H}' = 0 \quad \left(\text{since } \frac{\partial \vec{D}'}{\partial t} = 0\right)$

$\vec{B}' = \mu \vec{H}' = \mu_0 (\vec{H}' + \vec{M}')$

$\Rightarrow$ let $\vec{H}' = -\vec{\nabla} \psi$ (magnetic scalar potential)

$$\Rightarrow \vec{\nabla} \cdot \vec{B}' = \mu_0 \left(\vec{\nabla} \cdot \vec{H}' + \underbrace{\vec{\nabla} \cdot \vec{M}'}_{\substack{\| \\ 0}}\right) = -\mu_0 \nabla^2 \psi = 0.$$

as $\vec{M}' = \text{const}$

$\Rightarrow$ solve Laplace eqn

$$\psi(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + B_l r^{-(l+1)}\right) P_l(\cos\theta)$$

Outside $(r > R)$: expect field of a dipole

$$\Rightarrow \psi_>(r, \theta) = \frac{B_1}{r^2} P_1(\cos\theta) = \frac{B_1}{r^2} \cos\theta$$

Inside $(r < R)$: $\psi_<(r, \theta) = A_1 r P_1(\cos\theta) = A_1 r \cos\theta = A_1 z$

Note: fictitious magnetic surface charge density $\sigma = \vec{M}' \cdot \hat{n} = \vec{M}' \cdot \hat{r}$

$= M' \cos\theta$

Potential ψ is <u>continuous</u> $\Rightarrow$ $A_1 R\cos\theta = \frac{B_1}{R^2}\cos\theta$

$$B_1 = A_1 R^3.$$

$$\psi_< = A_1 z \Rightarrow \vec{H}'_{inside} = -\vec{\nabla}\psi = -A_1\hat{z}$$

$\vec{B}\cdot\hat{n} = \vec{B}\cdot\hat{r}$ is continuous ($\vec{\nabla}\cdot\vec{B} = 0$ always)

$$\Rightarrow \underbrace{\mu_0\left(\vec{H}' + \vec{M}'\right)\cdot\hat{r}\Big|_{r=R}}_{inside} = \underbrace{\mu_0\,\vec{H}'\cdot\hat{r}\Big|_{r=R}}_{outside}$$

$$\left(-A_1 + M'\right)\underbrace{\hat{z}\cdot\hat{r}}_{\cos\theta} = -\vec{\nabla}\psi_>\cdot\hat{r} = -\frac{\partial\psi_>}{\partial r}\Big|_{r=R} = \frac{2B_1}{R^3}\cos\theta$$

$$\Rightarrow -A_1 + M' = \frac{2A_1R^3}{R^3} \Rightarrow A_1 = \frac{1}{3}M', \quad B_1 = \frac{1}{3}M'R^3$$

$$\Rightarrow \begin{cases} \vec{H}'_{inside} = -\frac{1}{3}M'\hat{z} = -\frac{1}{3}\vec{M}' \\ \vec{B}'_{inside} = \mu_0\left(\vec{H}' + \vec{M}'\right) = \frac{2}{3}\mu_0\vec{M}' \end{cases}$$

$$\vec{H}'_{outside} = -\vec{\nabla}\left(\frac{\frac{1}{3}M'R^3\cos\theta}{r^2}\right) = -\left(\hat{r}\frac{\partial}{\partial r} + \frac{\hat{\theta}}{r}\frac{\partial}{\partial\theta}\right)\cdot$$

$$\cdot\,\frac{1}{3}M'R^3\frac{\cos\theta}{r^2}$$

$$\vec{H}'_{outside} = \frac{2}{3}M'\left(\frac{R}{r}\right)^3\cos\theta\,\hat{r} + \frac{1}{3}M'\left(\frac{R}{r}\right)^3\sin\theta\,\hat{\theta}.$$

$$\vec{B}'_{outside} = \mu_0\vec{H}'_{outside} \quad (\text{as } \vec{M}_{outside} = 0)$$

In the lab frame (where the sphere is rotating) we need to Lorentz transform (see pp. 6–7 for example)

$$\vec{E}'_{\parallel} = \vec{E}_{\parallel} \quad , \quad \vec{E}'_{\perp} = \gamma\left(\vec{E}_{\perp} + \vec{v}\times\vec{B}\right) \Rightarrow \left(\text{for } \frac{v}{c} \ll 1,\ \gamma \simeq 1\right)$$

$$\vec{E}_{\perp} = \vec{E}'_{\perp} - \vec{v}\times\vec{B}'$$

Similarly, $\vec{D}'_{\perp} = \gamma\left(\vec{D}_{\perp} + \frac{1}{c^2}\vec{v}\times\vec{H}\right) \Rightarrow \vec{D}_{\perp} \simeq \vec{D}'_{\perp} - \frac{1}{c^2}\vec{v}\times\vec{H}'$

Therefore $\vec{E}_{inside} = -\vec{v}\times\vec{B}'_{inside} = -\frac{2}{3}\mu_0\,\vec{\omega}\times\vec{r}\times M'\hat{z} =$

$$\left(\vec{v} = \vec{\omega}\times\vec{r}\right) = -\frac{2}{3}\mu_0\,\omega M'\,\underbrace{\hat{z}\times\vec{r}\times\hat{z}}_{\vec{r} - \hat{z}(\vec{r}\cdot\hat{z}) = \; = x\hat{x} + y\hat{y} = \vec{\rho}}$$

$$\vec{E}_{inside} = -\frac{2}{3}\mu_0\,\omega M'\rho\,\hat{\rho}$$ ($\hat{\rho}$ = unit vector in x–y plane, cylindrical coordinates)

Similarly $\vec{D}_{inside} = -\frac{1}{c^2}\,\vec{\omega}\times\vec{r}\times\left(-\frac{1}{3}M'\hat{z}\right) =$

$$= \frac{\omega}{3c^2}M'\rho\,\hat{\rho}.$$

$$\Rightarrow \vec{P}_{inside} = \vec{D} - \varepsilon_0\vec{E} = \frac{\omega}{3c^2}M'\vec{\rho} + \frac{2}{3}\underbrace{\mu_0\varepsilon_0}_{1/c^2}\,\omega M'\vec{\rho}$$

$$= \frac{\omega}{c^2}M'\vec{\rho}.$$

Since $\vec{B}_{inside} = \vec{B}'_{inside}$ $\left(\text{since } \vec{B}_{\parallel} = \vec{B}'_{\parallel},\ \vec{B}_{\perp} = \underbrace{\gamma}_{1}\left(\vec{B}'_{\perp} - \frac{1}{c^2}\vec{v}\times\underbrace{\vec{E}'}_{0}\right)\right)$

$$\Rightarrow \vec{E}_{inside} + \vec{v}\times\vec{B}_{inside} = \vec{F}_{Lorentz} = 0.$$

Electric field outside: $\vec{\nabla}\cdot\vec{E}_{out} = 0$

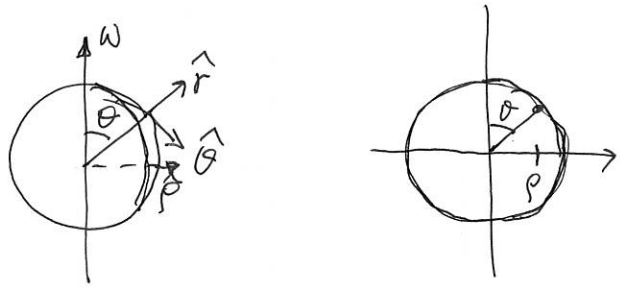
$$\vec{\nabla}\times\vec{E}_{out} = -\frac{\partial}{\partial t}\vec{B}_{out} = 0$$

($\vec{H}'_{out}$ is t-independent)

$$\Rightarrow \vec{E}_{out} = -\nabla\varphi_{out} \quad \text{and} \quad \nabla^2\varphi_{out} = 0.$$

Tangential components of $\vec{E}$ are continuous at $r=R$

$$\Rightarrow \hat{\theta}\cdot\vec{E}_{out} = \hat{\theta}\cdot\vec{E}_{in} = -\frac{2}{3}\mu_0\omega M'\rho\,\underbrace{\hat{\rho}\cdot\hat{\theta}}_{\cos\theta}\Big|_{r=R}$$



$$\rho\big|_{r=R} = R\sin\theta$$

$$\Rightarrow \vec{E}_{out}(r=R) = -\frac{2}{3}\mu_0\omega M' R\sin\theta\cos\theta\,\hat{\theta}$$

+ components along $\hat{r}$, $\hat{\varphi}$.

But $\varphi_{out}(r,\theta) = \sum_{l=0}^{\infty}\left(a_l r^l + b_l r^{-(l+1)}\right)P_l(\cos\theta)$

$$\vec{E}_{out} = -\vec{\nabla}\varphi_{out} = -\left(\frac{\partial}{\partial r}\varphi_{out}\,\hat{r} + \frac{1}{r}\frac{\partial\varphi_{out}}{\partial\theta}\hat{\theta}\right)$$

$$\Rightarrow \frac{\partial\varphi_{out}}{\partial\theta} \sim \cos\theta\sin\theta \Rightarrow \varphi_{out} \sim \cos^2\theta$$

$$\Rightarrow \varphi_{out} \sim P_{l=2}(\cos\theta) = \frac{1}{2}(3\cos^2\theta - 1)$$

$$\Rightarrow a_{l=2} = 0, \quad b_{l=2} \neq 0$$

$$\Rightarrow \varphi_{out}(r,\theta) = \frac{b_2}{r^3}P_2(\cos\theta)$$

$$\Rightarrow \vec{E}_{out}(r,\theta) = +\frac{3b_2}{r^4} P_2(\cos\theta)\,\hat{r} - \frac{1}{r}\frac{b_2}{r^3}\underbrace{\frac{\partial}{\partial\theta}\left(P_2(\cos\theta)\right)}_{-3\cos\theta\sin\theta}\hat{\theta}$$

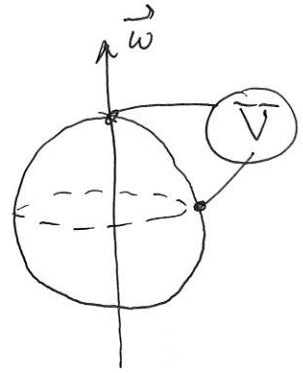
$$= \frac{3b_2}{r^4}\frac{1}{2}\left(3\cos^2\theta - 1\right)\hat{r} + \frac{3b_2}{r^4}\cos\theta\sin\theta\,\hat{\theta}$$

$$\Rightarrow \frac{3b_2}{R^4} = -\frac{2}{3}\mu_0\omega M' R \Rightarrow b_2 = -\frac{2}{9}\mu_0\omega M' R^5$$

$$\Rightarrow \vec{E}_{out}(r,\theta) = -\frac{\mu_0\omega M' R^5}{3r^4}\left[\left(3\cos^2\theta - 1\right)\hat{r} + 2\sin\theta\cos\theta\,\hat{\theta}\right]$$

Note that $\vec{E}_{out} \sim \frac{1}{r^4} \Rightarrow$ electric quadrupole.

$$\varphi_{out}(r,\theta) = -\frac{1}{9}\mu_0\omega M' R^5\,\frac{3\cos^2\theta - 1}{r^3}.$$



there is potential difference between points on the sphere with different latitudes.

Thus we found
$$\begin{cases} \vec{E}_{in} = -\frac{2}{3}\mu_0 \omega M' \rho \hat{\rho} \\ \vec{E}_{out} = -\frac{\mu_0 \omega M' R^5}{3 r^4}\left((3\cos^2\theta - 1)\hat{r} + 2\sin\theta\cos\theta\,\hat{\theta}\right) \end{cases}$$

$$\vec{\nabla}\cdot\vec{E}_{in} = \frac{\rho}{\varepsilon_0} \Rightarrow \rho(r,\theta) = -\frac{2}{3}\mu_0\varepsilon_0\,\omega M' \underbrace{\vec{\nabla}\cdot(\rho\hat{\rho})}_{\frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho^2) = 2} = -\frac{4\omega M'}{3c^2}$$

$\Rightarrow$ Total charge in the bulk of the sphere
$$Q_{bulk} = \frac{4\pi R^3}{3}\cdot\rho = -\frac{16\pi\,\omega M' R^3}{9c^2}.$$

<u>Surface charge density</u> $\frac{\sigma}{\varepsilon_0} = \left(\vec{E}_{out} - \vec{E}_{in}\right)\cdot\hat{r}$

$$\vec{\rho}\cdot\hat{r}\Big|_{r=R} = \left(\vec{r} - \hat{z}(\vec{r}\cdot\hat{z})\right)\cdot\hat{r}\Big|_{r=R} =$$
$$= R - R\cos^2\theta = R\sin^2\theta.$$

$$\Rightarrow \sigma = -\mu_0\varepsilon_0\,\omega M' R\left(\frac{3\cos^2\theta - 1}{3} - \frac{2}{3}\sin^2\theta\right)$$
$$= -\frac{\omega M' R}{c^2}\left(\cos^2\theta - \frac{1}{3} - \frac{2}{3}(1-\cos^2\theta)\right) =$$
$$= -\frac{\omega M' R}{c^2}\left(-1 + \frac{5}{3}\cos^2\theta\right) = \frac{\omega M' R}{c^2}\left(1 - \frac{5}{3}\cos^2\theta\right)$$

The total charge on the surface of the sphere is
$$Q_{surf} = \int da\,\sigma = R^2\int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta\left(1 - \frac{5}{3}\cos^2\theta\right)\frac{\omega M' R}{c^2}$$
$$= \frac{\omega M' R^3}{c^2}\cdot 2\pi\cdot\int_{-1}^{1} dx\left(1 - \frac{5}{3}x^2\right) = \frac{\omega M' R^3}{c^2}\cdot 2\pi\cdot\frac{8}{9} = -Q_{bulk}$$

$x = \cos\theta \qquad 2\left(1 - \frac{5}{9}\right) = \frac{8}{9} \qquad \Rightarrow \boxed{\text{total charge} = 0\,!!!}$

$$L^{\alpha}{}_{\beta} = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} ; \quad F^{\alpha\beta} = \begin{pmatrix} 0 & -\frac{E_x}{c} & -\frac{E_y}{c} & -\frac{E_z}{c} \\ \frac{E_x}{c} & 0 & -B_z & B_y \\ \frac{E_y}{c} & B_z & 0 & -B_x \\ \frac{E_z}{c} & -B_y & B_x & 0 \end{pmatrix}$$

$$F'^{\mu\nu} = L^{\mu}{}_{\alpha} L^{\nu}{}_{\beta} F^{\alpha\beta}$$

$$F'^{01} = L^{0}{}_{\alpha} L^{1}{}_{\beta} F^{\alpha\beta} = L^{0}{}_{\alpha} \left(L^{1}{}_{0} F^{\alpha 0} + L^{1}{}_{1} F^{\alpha 1} + L^{1}{}_{2} F^{\alpha 2} + L^{1}{}_{3} F^{\alpha 3} \right)$$

$$= L^{0}{}_{\alpha} \left(-\beta\gamma F^{\alpha 0} + \gamma F^{\alpha 1} \right) = -\beta\gamma \left(L^{0}{}_{0} \cancel{F^{00}} + L^{0}{}_{1} F^{10} + \cancel{L^{0}{}_{2}} F^{20} + \cancel{L^{0}{}_{3}} F^{30} \right) + \gamma \left(L^{0}{}_{0} F^{01} + L^{0}{}_{1} \cancel{F^{11}} + \cancel{L^{0}{}_{2}} F^{21} + \cancel{L^{0}{}_{3}} F^{31} \right) = -\beta\gamma \left(-\beta\gamma F^{10} \right) + \gamma \left(\gamma F^{01} \right) = \gamma^2 (1-\beta^2) F^{01} = F^{01}$$

$L^{1}{}_{0} = -\beta\gamma$

$L^{1}{}_{1} = \gamma$

$$F'^{01} = F^{01} \Rightarrow E'_x = E_x$$

$L^{0}{}_{1} = -\beta\gamma$

$L^{0}{}_{0} = \gamma$

$$F'^{02} = L^{0}{}_{\alpha} L^{2}{}_{\beta} F^{\alpha\beta} = L^{0}{}_{\alpha} \underbrace{L^{2}{}_{2}}_{=1} F^{\alpha 2} = L^{0}{}_{0} F^{02} + L^{0}{}_{1} F^{12} + L^{0}{}_{2} \cancel{F^{22}} + \cancel{L^{0}{}_{3}} F^{32}$$

the only $\neq 0$ element $L^{2}{}_{2}$

$$= \gamma F^{02} - \beta\gamma F^{12}$$

$$-\frac{1}{c} E'_y = \gamma \left(-\frac{1}{c} E_y \right) - \beta\gamma \left(-B_z \right)$$

$$\Rightarrow E'_y = \gamma E_y - \gamma v B_z = \gamma \left(E_y - v B_z \right)$$

$$F'^{12} = L^{1}{}_{\alpha} L^{2}{}_{\beta} F^{\alpha\beta} = L^{1}{}_{\alpha} \underbrace{L^{2}{}_{2}}_{=1} F^{\alpha 2} = L^{1}{}_{0} F^{02} + L^{1}{}_{1} F^{12} + L^{1}{}_{2} \cancel{F^{22}} + L^{1}{}_{3} F^{32} = -\beta\gamma \left(-\frac{E_z}{c} \right) + \gamma \left(-B_z \right)$$

$-B'_z =$

$$\Rightarrow B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

Thus
$$\begin{cases} E'_x = E_x & \checkmark \\ E'_y = \gamma\left(E_y - v B_z\right) & \checkmark \\ E'_z = \gamma\left(E_z + v B_y\right) & \\ B'_x = B_x & \\ B'_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right) & \\ B'_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right) & \checkmark \end{cases}$$

$$\vec{v}\times\vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ v_x & 0 & 0 \\ B_x & B_y & B_z \end{vmatrix} = -v_x B_z \hat{y} + v_x B_y \hat{z}$$

$$\vec{E}'_\perp = \gamma\left(\vec{E}_\perp + \vec{v}\times\vec{B}\right)$$

$$\vec{B}'_\parallel = \vec{B}_\parallel$$

$$\vec{B}'_\perp = \gamma\left(\vec{B}_\perp - \frac{1}{c^2}\vec{v}\times\vec{E}\right)$$

Therefore tensor
$$H^{\mu\nu} = \begin{pmatrix} 0 & -cD_x & -cD_y & -cD_z \\ cD_x & 0 & -H_z & H_y \\ cD_y & H_z & 0 & -H_x \\ cD_z & -H_y & H_x & 0 \end{pmatrix}$$

will lead to (via $\frac{\vec{E}}{c} \to c\vec{D}$, $\vec{B} \to \vec{H}$ replacement)

$$D'_x = D_x$$

$$\frac{E'_y}{c} = \gamma\left(\frac{E_y}{c} - \beta B_z\right) \to cD'_y = \gamma\left(cD_y - \beta H_z\right)$$

$$D'_y = \gamma\left(D_y - \frac{v}{c^2}H_z\right)$$

$$D'_z = \gamma\left(D_z + \frac{v}{c^2}H_y\right) \longrightarrow \vec{D}'_\perp = \gamma\left(\vec{D}_\perp + \frac{1}{c^2}\vec{v}\times\vec{H}\right)$$

$$H'_x = H_x$$

$$\frac{E_z}{c} \to cD_z$$

$$H'_y = \gamma\left(H_y + \frac{v}{c}(cD_z)\right) = \gamma\left(H_y + vD_z\right)$$

$$H'_z = \gamma\left(H_z - vD_y\right)$$